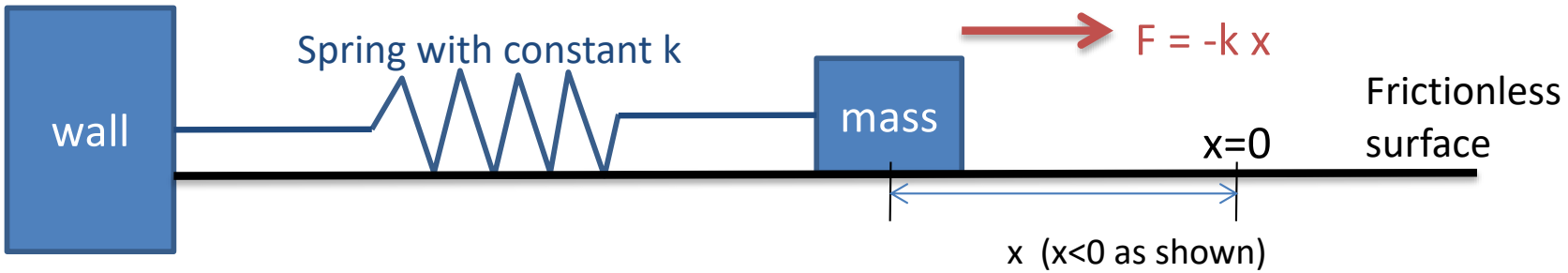
SIMPLE HARMONIC MOTION

Hooke's Law



- As you push a spring farther from equilibrium, it exerts a greater force.
- Letting $x = 0$ at equilibrium,

$$F = -kx$$

- Typical for many other systems, such as rigid beams, chemical bonds, voltage in an LC resonant circuit,…

Simple Harmonic Motion

Hooke's Law: $\vec{F} = -k\vec{x}$

Plus Newton's 2nd law: $\vec{F} = m\vec{a} = m\frac{d^2\vec{x}}{dt^2}$

Gives a 2nd order Ordinary Differential Equation (ODE):

$$m\frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

NOTE: You CANNOT use formulas like $v = at$ or $x = \frac{1}{2}at^2$ for SHM

These are valid ONLY for constant acceleration!

- Solution depends on the sign of k/m
  - For a mass on a spring, k and m are positive
  - Restrict to case k/m>0 by writing $\omega^2 = k/m$
- $\frac{d^2x}{dt^2} + \omega^2 x = 0$ has a general solution:

$$x(t) = A\cos(\omega t + \phi_0)$$

  - This can be verified by substitution
- 2nd order ODE, so two arbitrary constants A, ϕ_0

- A is a constant → **amplitude**
- ω ("omega") is a constant → **angular frequency**
  - Units: radians per second
  - Related to the frequency f (in Hz) as: $\omega = 2\pi f$
  - Related to the period T (in s) as: $f = 1/T$
    - So that $\omega = 2\pi/T$
- ϕ_0 ("phi naught") is a constant →**phase angle**
  - Initial condition – shifts curve left/right

Clarification on using $\cos(\omega t + \phi_0)$

We saw the solution to the ODE was in terms of $e^{j\omega t}$ and $e^{-j\omega t}$, or equivalently in terms of

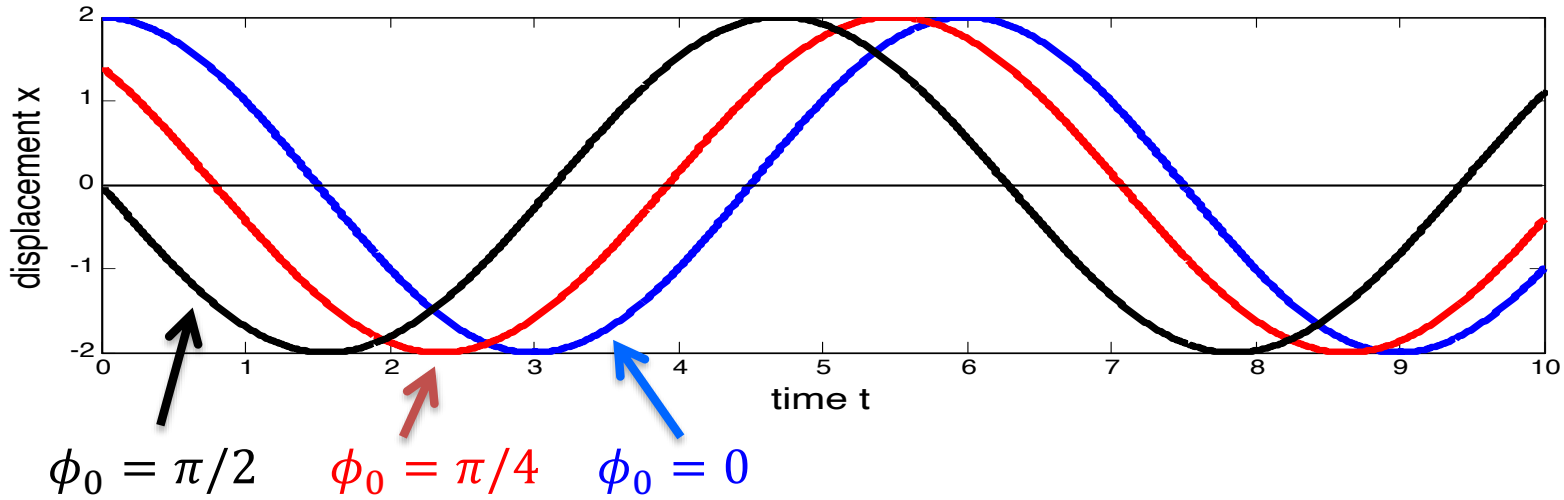
$$\cos \omega t = \frac{1}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$$

$$\sin \omega t = \frac{1}{2}\left(e^{j\omega t} - e^{-j\omega t}\right)$$

so why then do we use $\cos(\omega t + \phi_0)$?

→ Doc cam

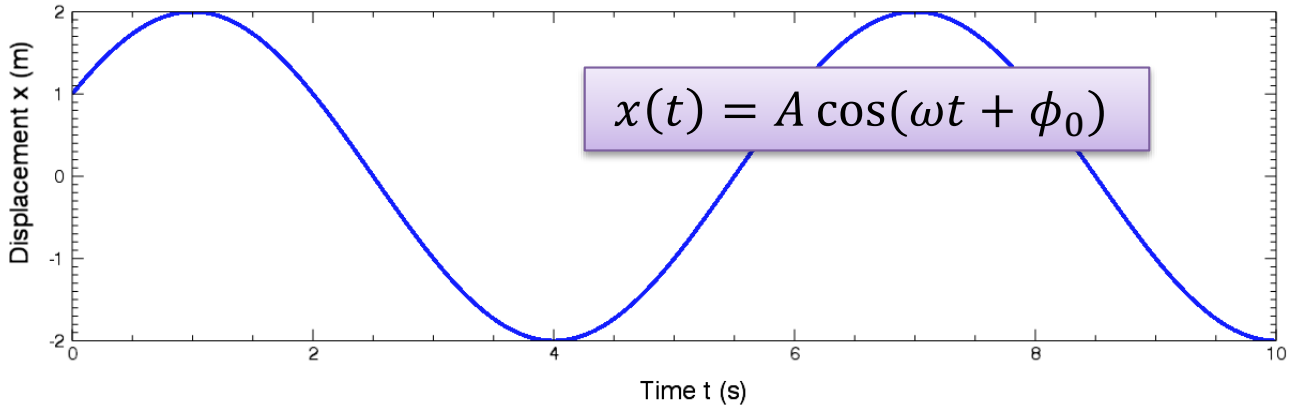
$$x(t) = A\cos(\omega t + \phi_0)$$



- ϕ_0>0 shifts curve to the left
  - ϕ_0 <0 shifts curve to the right
- $\cos\left(\omega t + \frac{\pi}{2}\right) = -\sin(\omega t)$

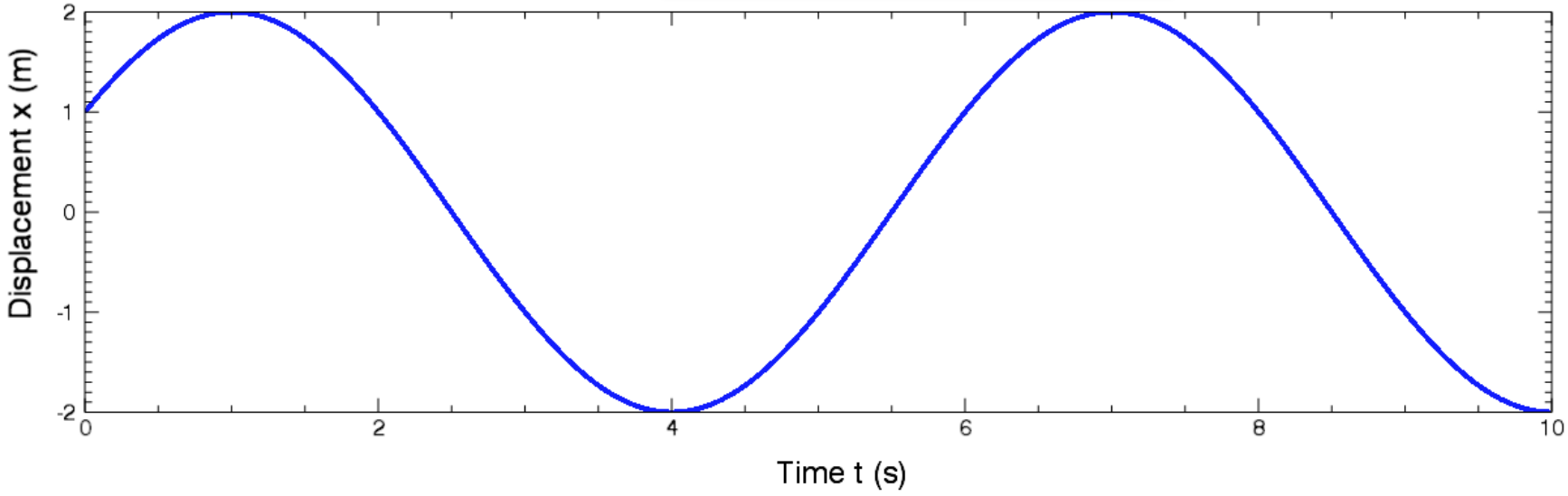
EXAMPLE

Find A, ω



Find A, ω

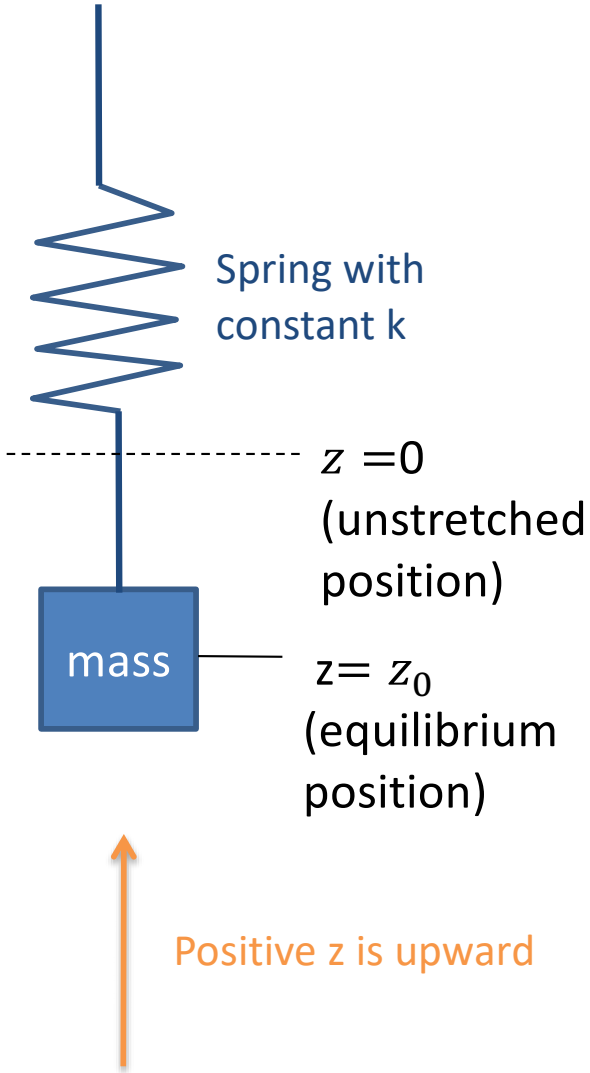
$$x(t) = A\cos(\omega t + \phi_0)$$



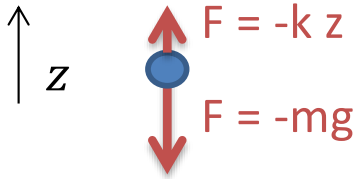
Amplitude: $\boldsymbol{A}$**= 2 m**

Angular frequency: Period T = 6 s. Using $\omega = \frac{2\pi}{T}$: $\boldsymbol{\omega = \pi/3}$ **radians/s**

Motion of a Vertical Spring



Free-body diagram:



$$F = -kz - mg$$

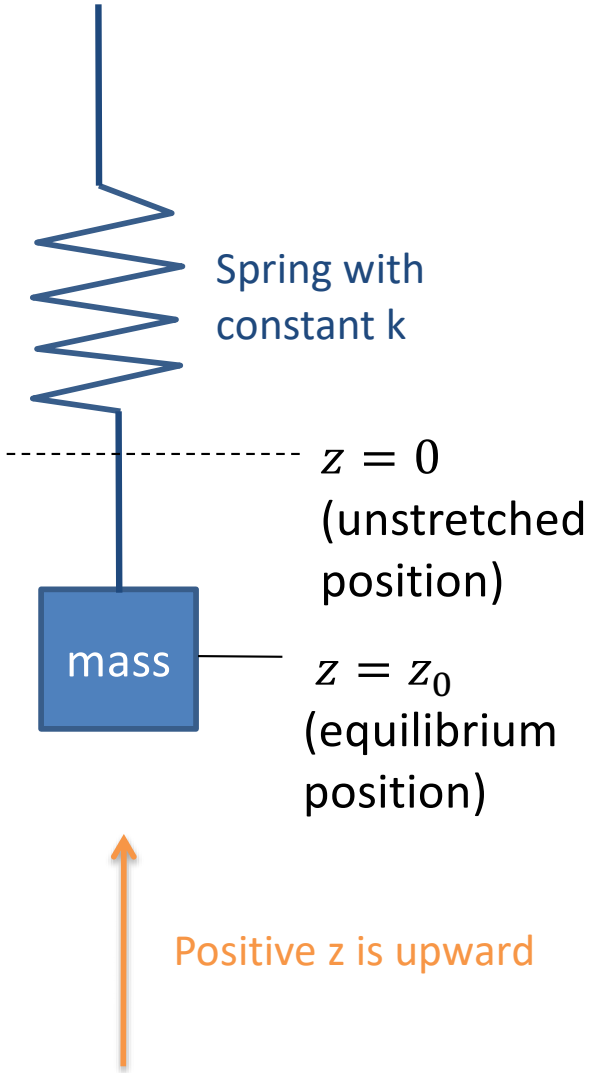
Equilibrium is when $F = 0$.

Let z_0 be the distance between the unstretched and equilibrium positions:

$$0 = -kz_0 - mg$$

$$kz_0 = -mg$$

$$z_0 = -mg/k$$



Then: $F = -kz - mg$

becomes $F = -k(z - z_0)$

If we now let $y = z - z_0$

then we have the SHM force law:

$$F = -ky$$

Motion is SHM:

$$y(t) = A\cos(\omega t + \phi_0)$$

Or $z(t) = A\cos(\omega t + \phi_0) + z_0$



Example: baby in a jumper

Mass: m = 10 kg
Equilibrium displacement: $z_0 = -10\ \mathrm{cm}$

Find frequency and period of the motion

Equilibrium displacement z_0 is -10cm, so using

$$z_0 = \frac{-mg}{k}$$

$$k = \frac{-mg}{z_0} = \frac{-10kg \cdot 9.81\, m/s^2}{-10cm} = 981\, N/m$$

The angular frequency of oscillation is

$$\omega = \sqrt{k/m} = \sqrt{(981\, N/m)/10kg} = 9.9\, rad/s$$

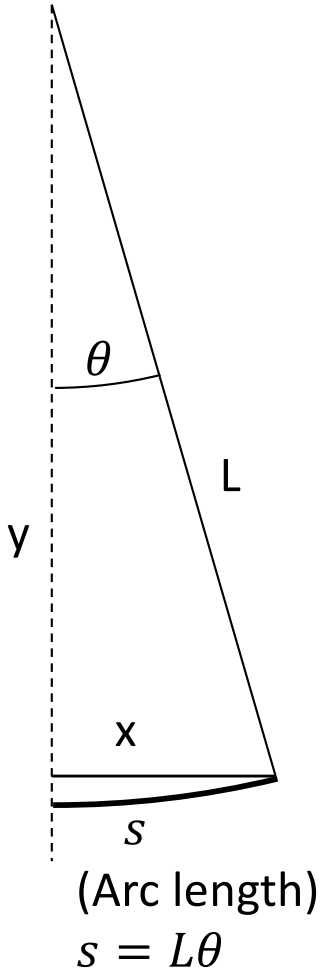
The period is

$$T = \frac{1}{f} = \frac{2\pi}{\omega} = \mathbf{0.6\ s}$$

We can check this experimentally

THE SIMPLE PENDULUM

The Small Angle Approximation



We will use this a lot in this course:

$$\sin\theta = \frac{x}{L}$$

$$\sin\theta \approx \frac{s}{L}$$

Also:

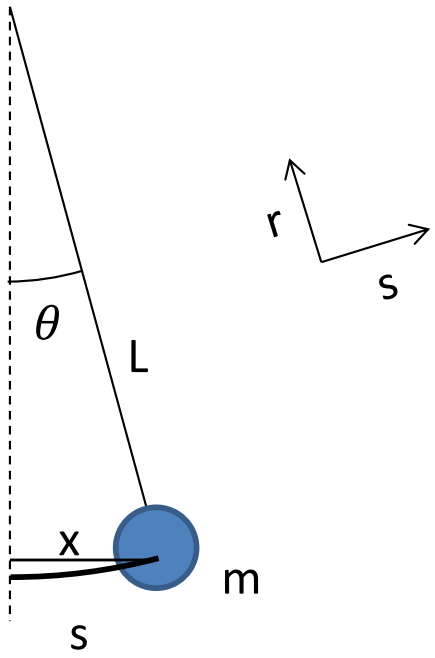
$$\boxed{\sin\theta \approx \theta}$$

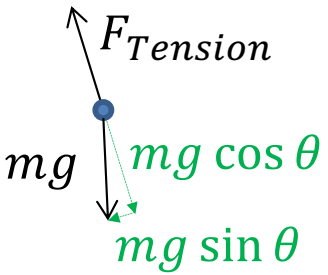
$$\boxed{\tan\theta \approx \theta}$$

Error is < few % for about $|\theta| < 15°$

Simple Pendulum

(Small Angle Solution)





- Net force in s direction:

$$F_s = -mg\sin(\theta)$$

$$F_s = -mg(x/L)$$

$$F_s \approx -mg(s/L)$$

Compare to $F = -kx$ with $\omega^2 = k/m$

- Resulting motion: SHM with

$$\omega^2 = g/L$$